Name_____ Date_____ Period_____

LESSON 5-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Trapezoids by Decomposing

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles.

REMEMBER

- **Decompose** a trapezoid into a rectangle and one or two triangles.
- Find the area of each piece, then **add** the areas.
- When you cut off a rectangle, the triangle bases together equal the **difference** of the two parallel sides.

WORKED EXAMPLE

An isosceles trapezoid has parallel sides 6 m and 12 m and height 5 m. Find the area.

Step 1 Rectangle: 6 × 5 = 30

Step 2 Triangle bases: (12 − 6) ÷ 2 = 3 m each

Step 3 Triangles: 2 × (½ × 3 × 5) = 15

Step 4 Add: 30 + 15 = 45

Answer **45 m²**

WORD BANK

trapezoid – 4 sides, one pair of parallel sides

parallel – always the same distance apart

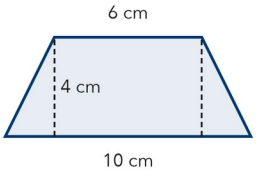
decompose – cut into smaller shapes

triangle area – ½ × base × height

PRACTICE

show your work · label units

1. The trapezoid is cut into a rectangle and two matching triangles.



Step 1 First find the rectangle: top × height.

Step 2 Then find one triangle: ½ × base × height.

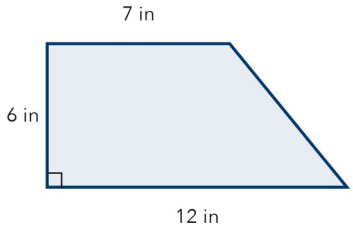
Step 3 Add the rectangle and both triangles.

a. Area of the rectangle _____ cm²

b. Area of one triangle _____ cm²

Area of trapezoid_____ cm²

2. Decompose the trapezoid into a rectangle and one triangle. Find its area.



Step 1 Rectangle: short side × height.

Step 2 Triangle base: 12 − 7.

Step 3 Add the two areas.

Show your work.

Answer_____ in²

3. **Skate Ramp.** A skate ramp side is a trapezoid with one **square corner**. The parallel sides are 3 ft and 8 ft. The height is 4 ft. Cut it into a rectangle and a triangle. Find the area.

Step 1 Rectangle: 3 × 4.

Step 2 Triangle base: 8 − 3.

Step 3 Add the two areas.

Show your work.

Answer_____ ft²

4. **Table Top.** A table top is a **trapezoid**. The parallel sides are 5 ft and 11 ft. The height is 3 ft. Cut it into a rectangle and two **matching** triangles. Find the area.

Step 1 Find the rectangle.

Step 2 Split 11 − 5 into two equal triangle bases.

Step 3 Add all three pieces.

Answer_____ ft²

5. **Find the Error.** A trapezoid has parallel sides 9 cm and 15 cm and height 4 cm. Tomás says each of its 2 matching triangles has a base of 6 cm. Explain his mistake and find the area.

Frame *Tomás is wrong because the two triangle bases together are ___, so each one is ___ cm.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a trapezoid shape at home or outside (a lampshade side, a bag, a roof). Show someone how to cut it into a rectangle and triangles.

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The Trapezoid Area Formula

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a trapezoid using the trapezoid area formula.

REMEMBER

- **A = ½ × (b1 + b2) × h**. The bases b1 and b2 are the **parallel** sides.
- Add the bases first, then multiply by the height *h*, then take half.

WORKED EXAMPLE

Parallel sides 7 cm and 11 cm, height 6 cm. Find the area.

Step 1 Add the bases: 7 + 11 = 18

Step 2 Multiply by the height: 18 × 6 = 108

Step 3 Take half: 108 ÷ 2 = 54

Answer **54 cm²**

WORD BANK

base – one of the 2 parallel sides

height – straight distance between the bases

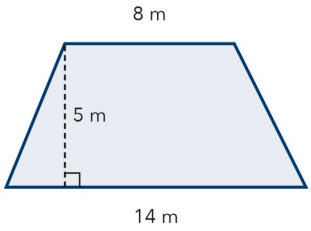
formula – A = ½ × (b1 + b2) × h

sum – the answer when you add

PRACTICE

show your work · label units

1. Find the area of the trapezoid.



Step 1 Add the bases: 8 + 14.

Step 2 Multiply by the height.

Step 3 Take half.

Show your work.

Answer_____ m²

2. Find the area of each trapezoid with bases b1 and b2 and height *h*.

a. 4 ft, 10 ft; h = 6 ft _____ ft²

b. 5 in, 9 in; h = 8 in _____ in²

c. 6 m, 12 m; h = 3 m _____ m²

d. 3 cm, 7 cm; h = 9 cm _____ cm²

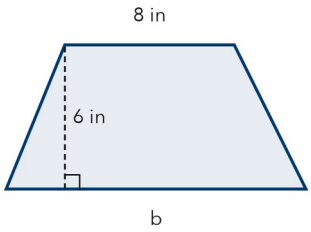
3. Garden Plot. A garden is a **trapezoid**. The parallel sides are 16 ft and 24 ft. The height is 10 ft. One bag of mulch covers 25 ft². How many bags do you need?

Step 1 Find the area with the formula.

Step 2 Then divide the area by 25.

Answer_____ bags

4. Missing Base. The area of the trapezoid is 60 in². Find the missing base *b*.



Step 1 Double the area: 2 × 60.

Step 2 Divide by the height. This gives b1 + b2.

Step 3 Subtract the base you know.

Show your work.

Answer_____ in

TEST PRACTICE

choose the best answer

5. A trapezoid has parallel sides of 6 cm and 10 cm and a height of 5 cm. What is its area?

A 21 cm² **B** 40 cm² **C** 80 cm² **D** 300 cm²

6. Which expression finds the area of a trapezoid with bases 3 m and 9 m and height 4 m?

Step 1 The bases are added first, inside ( ).

Step 2 Then multiply by the height and take half.

A ½ × 3 × 9 × 4 **B** (3 + 9) × 4 **C** ½ × (3 + 9) × 4 **D** ½ × 3 + 9 × 4

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Measure a trapezoid shape at home, like the side of a bag. Estimate its area with the formula.